01 / THE QUESTION

Are we moving in the right direction?

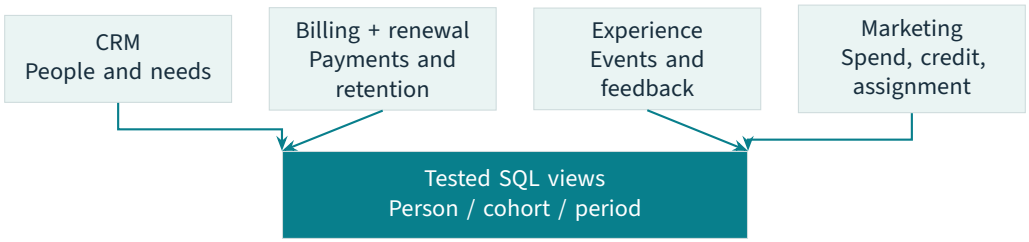
One subscription business. Eight separate exports. A diagnosis that connects customer experience, acquisition and durable contribution.

Latest observed month
2,576
retained subscribers

Latest observed month
$179,566
net revenue

Latest observed month
$85,410
contribution before acquisition spending

The Universal Customer View connects meanings and keys



Interactive source-system and customer-journey diagrams are generated with the bundled, pinned Archify package. This report schematic is an original companion view of the same data domains.

UCV category	What the exported records mean
Identifiers	`person_id` connects documented records. Transaction, period and event IDs identify different objects.
Relationship	Subscription period, tier and renewal outcome describe the arrangement and when it applies.
Engagement	Payments, product actions and support contacts are observed events, not inferred motives.
Perceptions	Survey responses describe the respondents' evaluations. No response remains missing.

What would count as progress?

Compare customers at the same age since first payment, inspect contribution after acquisition spending, then connect joining cohorts to the firm's calendar-time results. The experiment asks a separate question: did the tested campaign add value relative to its control?

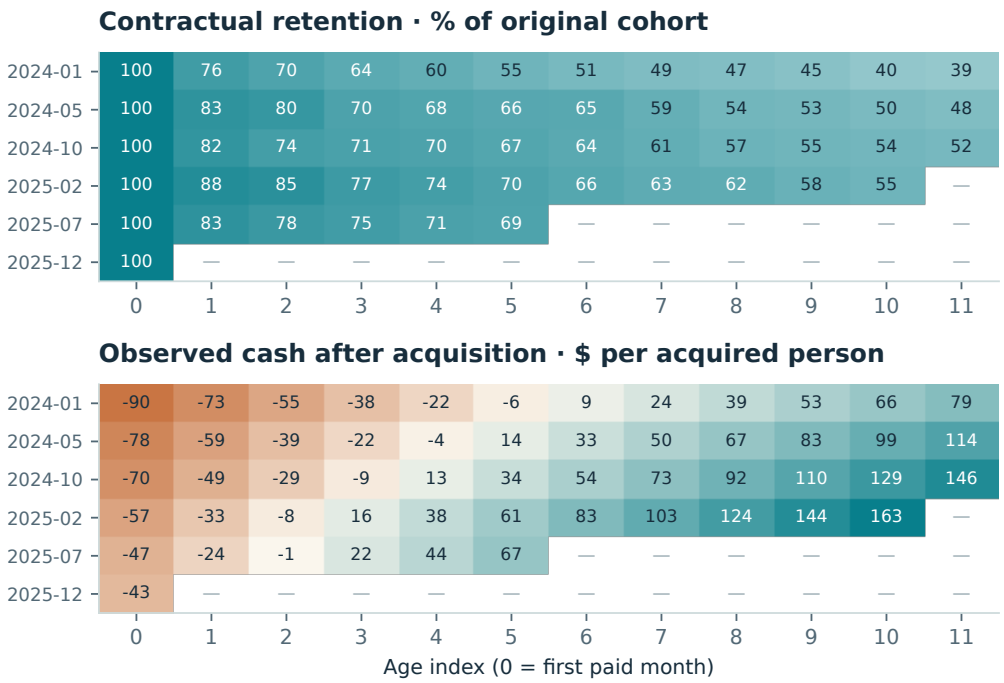
A larger business is not sufficient evidence of better customer economics. We need age-aligned cohorts, a stated cost scope, and a credible comparison for marketing effects.

Data as of 2026-01-01T00:00:00Z. All records are original synthetic teaching data. Forecast horizon: 12 monthly periods from acquisition. Detailed file and SQL hashes are recorded in the report manifest.

02 / OBSERVED COHORTS

Start with a cohort, not a company average

A cohort contains people who first paid in the same month. Age index 0 is the first paid month; blank future cells are unobserved.



Source: saved SQL tables `cohort_totals.csv` and `cohort_cash_economics.csv`. The display selects joining cohorts for legibility; it does not interpolate missing future observations.

Read across, then compare down

Across a row, follow one joining cohort as it ages. Down an age column, compare different cohorts at the same stage of their relationship. A newest cohort's short history cannot be compared with an older cohort's accumulated total.

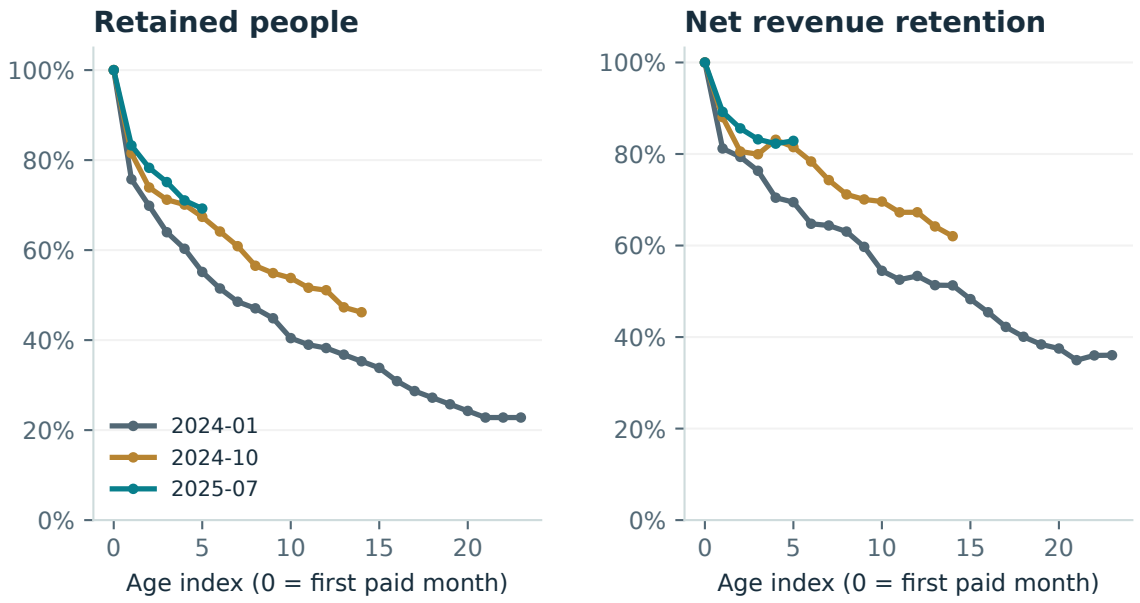
Quantity	Denominator and interpretation
Retained share	Retained subscriptions divided by the original acquired cohort; product inactivity is not cancellation.
Paying customers	People with a successful payment in the period; this is a purchase measure, not a substitute for contractual retention.
Cash after acquisition	Cumulative observed contribution less the cohort's tracked acquisition spending. Cash payback is the first nonnegative balance.

First-paid month is derived from complete payment history. CRM creation, product signup and marketing attribution are separate events. Refund rows reduce value; they do not create extra orders or new people.

03 / CUSTOMER DIFFERENCES

People and revenue can tell different stories

A falling customer count can coexist with stronger spending among those who remain. Interpret both curves before describing retention as good or bad.



Observed SQL ratios use original cohort membership for people and the cohort's first-period revenue for revenue retention. Selected cohorts are labeled; all plotted observations are saved SQL outputs.

Segmentation represents selected differences

The pre-acquisition need groups and age groups organize the same customers differently. The comparison below uses mature customers at a common six-month horizon. Smaller within-group variation can make a grouping more informative; it does not make everyone within a segment identical.

Grouping	Within-group variance	Groups	Smallest group
age band	4,873.1	2	1667
primary need	4,671.1	2	1566
region	4,873.3	2	1695

Use the journey to interpret, not invent, explanations

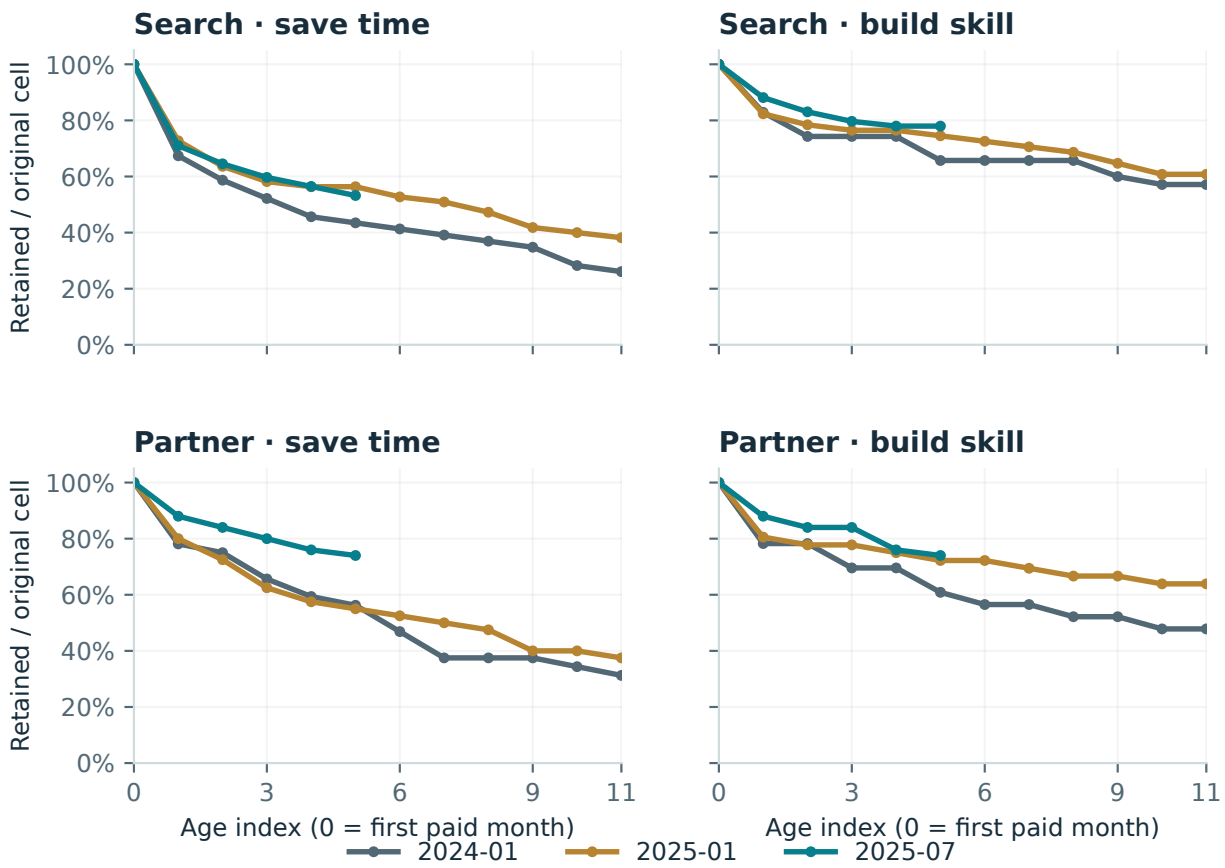
An onboarding event records an action. A survey rating records an expressed evaluation. Their timing can suggest a question worth investigating, but a customer path alone does not establish why someone renewed or left.

Joining month defines a cohort. Stated need or age defines a segment. Customer heterogeneity exists before we choose either representation.

04 / CHANNELS AND NEEDS

Check the trend within customer groups

Compare the same joining cohorts inside each channel and pre-acquisition need. The axes stay fixed across all four panels.



Source: `cohort_age_summary.csv`. Each curve divides retained customers by its original cohort–channel–need cell. Only observed ages appear; the July 2025 curves stop at age 5. No future zeros or filled-in retention paths are added.

What survives the breakdown?

At observed age index 5, July 2025 retains a larger share than January 2024 in 4 of the 4 channel-by-need cells. This supports investigating improvement within the groups, rather than attributing the whole aggregate change to a different mix. The curves show where that comparison holds and where later outcomes remain unobserved.

Use the differences to ask a better question

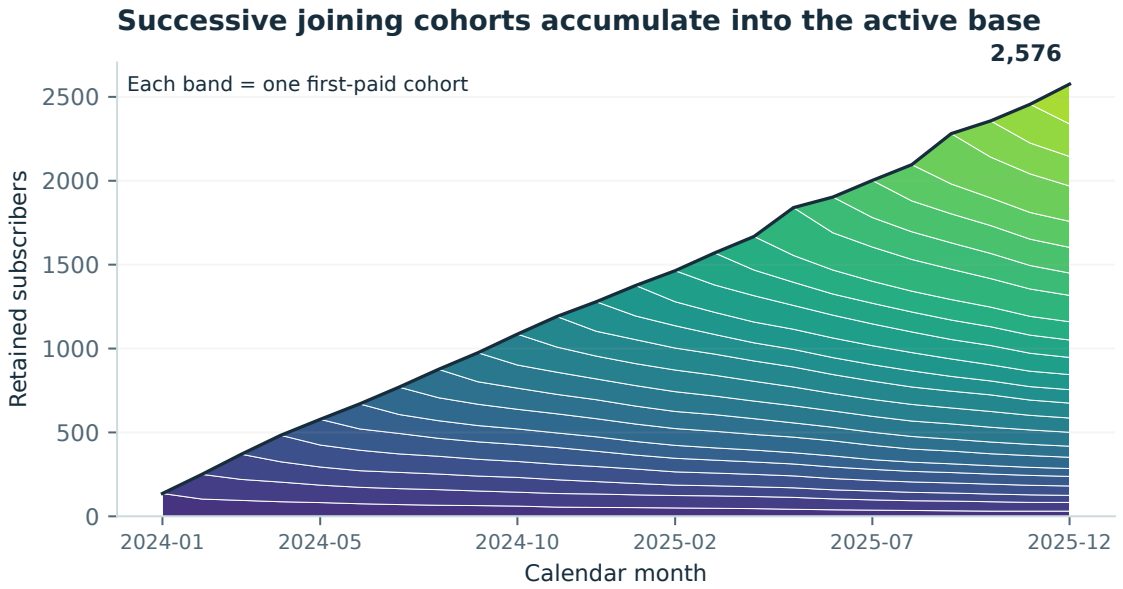
If a channel's aggregate curve changes, examine both its customer mix and the trajectories within the same needs. Different curves can guide a diagnostic question or a new test; they do not show that the acquisition channel caused the difference.

A cohort controls the joining clock. A segment organizes customer differences. Use both before explaining a company-wide trend.

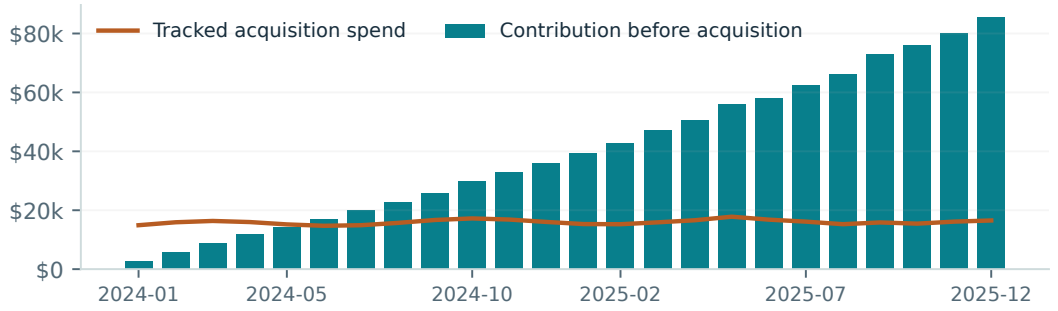
05 / COMPANY DYNAMICS

Cohort layers build the company over time

Each colored layer is a joining cohort. The firm grows when new customer inflows and retained older cohorts together outweigh departures.



Source: `company_month.csv` cohort layers and independently saved SQL company totals. Layer sums are checked against totals before plotting.



Separate scale from economic quality

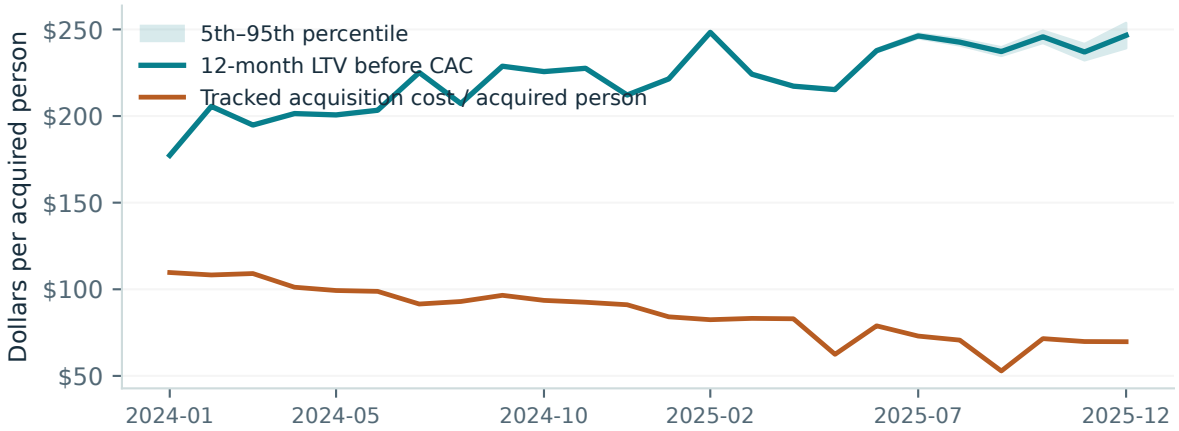
The upper figure shows active customers at each calendar month, not cumulative acquisitions. The lower figure compares contribution before acquisition spending with the spending incurred to acquire customers. A growing active base can still require substantial acquisition investment.

Contribution subtracts recorded variable costs. Fixed operating costs, financing and taxes are outside this dataset. These figures do not establish company profitability.

06 / FORECAST VALUE

Make the horizon explicit before comparing LTV

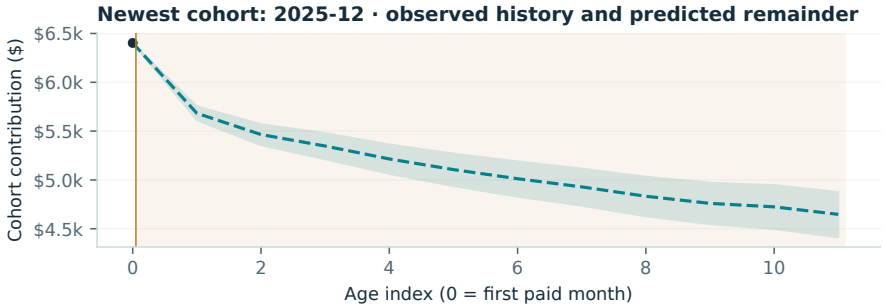
Observed contribution is retained. Only the unobserved remainder of the common 12-month horizon is forecast.



Bands are approximate 5th–95th posterior parameter percentiles. They exclude optimization, model-structure and future-economics uncertainty; they are not full predictive intervals.

$$\text{LTV}_{12} = \sum_{m=1}^{12} \frac{\mathrm{E}[c_m]}{(1.01)^m}, \qquad \text{net LTV}_{12} = \text{LTV}_{12} - \text{CAC}.$$

c_m is monthly contribution per original acquired customer, including zeros for those who have left. The first receipt is discounted at $m = 1$ (saved age index 0). Future net revenue and contribution shares of payment revenue use observed channel-by-need-by-age shares, with the same group's observed renewal ages (age>=1) as fallback. Shares are fixed assumptions, not causally identified or assigned posterior uncertainty.



Test prediction against outcomes we withheld

Withheld cells: 258. Retained-count mean absolute error: 1.09. Revenue weighted absolute percentage error: 5.5%. A separate training-only forecast checks complete 12-month values: LTV mean absolute error $6.72 versus $34.87 for its baseline. These tests evaluate prediction, not campaign effects. For the 2025-12 cohort, forecast 12-month net LTV is $176.83 per acquired person (5th–95th percentile: $169.46 to $183.95). Forecast share of LTV: 89%.

The shared joining-cohort trend may miss deterioration within a particular segment; inspect the separate recent-cohort fit diagnostic. Forecast LTV and historical average CAC do not identify the causal return on the next advertising dollar. Cash payback remains undiscounted.

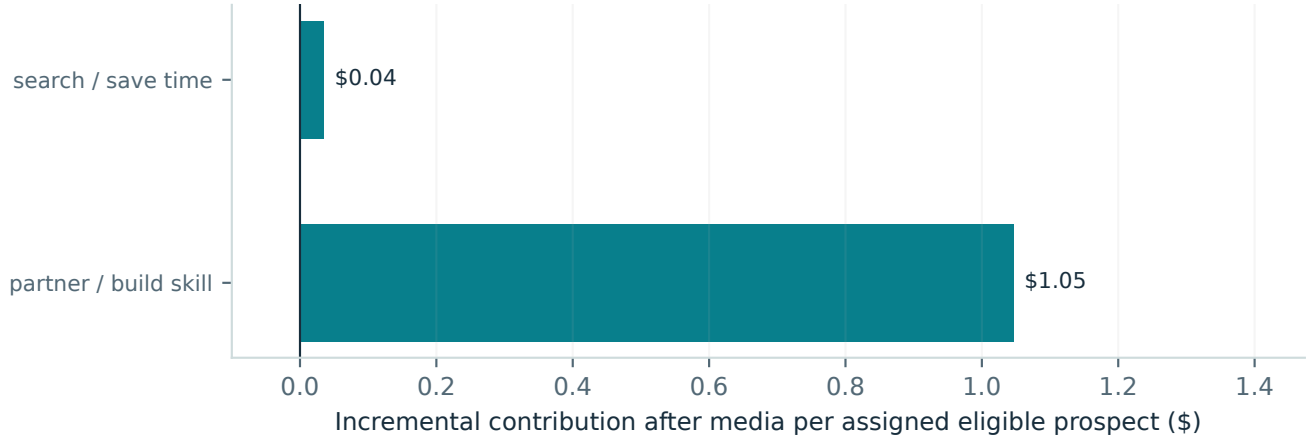
07 / RANDOMIZED EVIDENCE

A campaign must beat its no-campaign baseline

Start from every assigned eligible prospect, including people who never buy. Their zero outcomes belong in the denominator.

Test	Arm	Assigned	Buyers	Contribution	Media
search / save time	control	250	24	$551.60	$0.00
search / save time	treatment	250	51	$1,210.40	$650.00
partner / build skill	control	250	26	$660.15	$0.00
partner / build skill	treatment	250	67	$1,821.80	$900.00

Source: `experiment_summary.csv` and `experiment_effects.csv`. Each experiment keeps its own assignment roster, control and prespecified follow-up window. Organic purchases by control members remain control outcomes.



These are point estimates over the tested window, not certain rankings. The analysis tables separately report uncertainty for the conversion-rate difference.

Compare contribution using the same transformation

For each arm, net revenue less variable costs gives contribution. Compare arm means per eligible prospect; then subtract the treatment's additional media cost per assigned prospect. Purchases that would have occurred without the campaign are part of the baseline.

$$\Delta\text{contribution after media per prospect} = \frac{C_T - M_T}{N_T} - \frac{C_C - M_C}{N_C}.$$

The experimental window is not a lifetime horizon. Do not multiply a conversion lift by the average predicted LTV of observed buyers and call the result experimentally established lifetime incrementality.

08 / THE MANAGEMENT DECISION

Connect the evidence before choosing the next move

The task is to judge whether the business is improving and identify the next measurement or operating action.

Evidence	Earlier	Later	Interpretation
Sixth-month retained share	55.1%	69.2%	Observed at age index 5
Tracked cost per acquired person	$109.68	$72.99	Defined spending / all acquired people
Joining cohort	2024-01	2025-07	Latest cohort with six observed ages

A defensible conclusion

At the common sixth age month, retained share is higher: 55.1% for 2024-01 versus 69.2% for 2025-07. Tracked acquisition cost per acquired person is lower: $109.68 versus $72.99. This blended CAC includes all acquired people, including organic control buyers; paid-source CAC has a different denominator. These are descriptive comparisons, not campaign effects.

Learner judgment and next action

Judgment: Instructor reference build: no learner judgment has been recorded.

Recommended action: Choose the next action after reviewing the observed, forecast and experimental evidence.

Carry the decision into action

Responsibility	Output needed before acting
Marketing + Product	State the customer objective, eligible population and definition of success.
Engineering + Analytics	Preserve source meaning, the person key, observation cutoffs and tested SQL transformations.
Operations + Marketing	Verify the final recipients or rollout configuration and own the release decision.
Evaluation	Keep assignment and nonconverters; assess the outcome over the prespecified comparison window.

Decide from the evidence that exists. Name the uncertainty that could change the decision, and make the next action produce the missing evidence.

Reproducibility and sources

All numerical tables are saved SQL or model outputs. The chart manifest binds each figure to those files. The report manifest records file hashes, model metadata and the compiler. SQL scripts, editable LaTeX and original diagram sources accompany this PDF.

Conceptual references: Juan Orduz, *Revenue Retention and Customer Lifetime Value with NumPyro* (public walkthrough); Gil Bouhnick, *Cohort Analysis* (MobileSpoon, 2019). This lab uses original synthetic data and explicit contribution costs; it does not copy article images or treat an article's transaction commission as its own contribution margin. Archify 3.2.0, MIT, pinned source included with notices. See `docs/visual-sources.md` for links and license details.